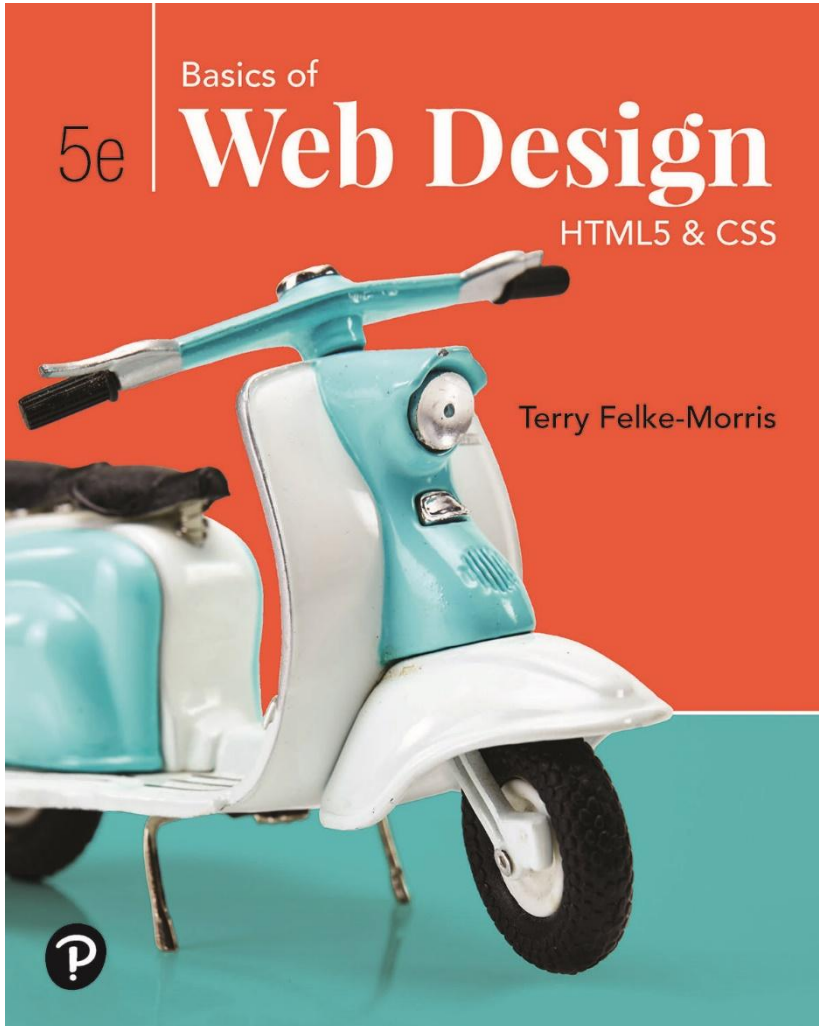


Basics of Web Design

Fifth Edition



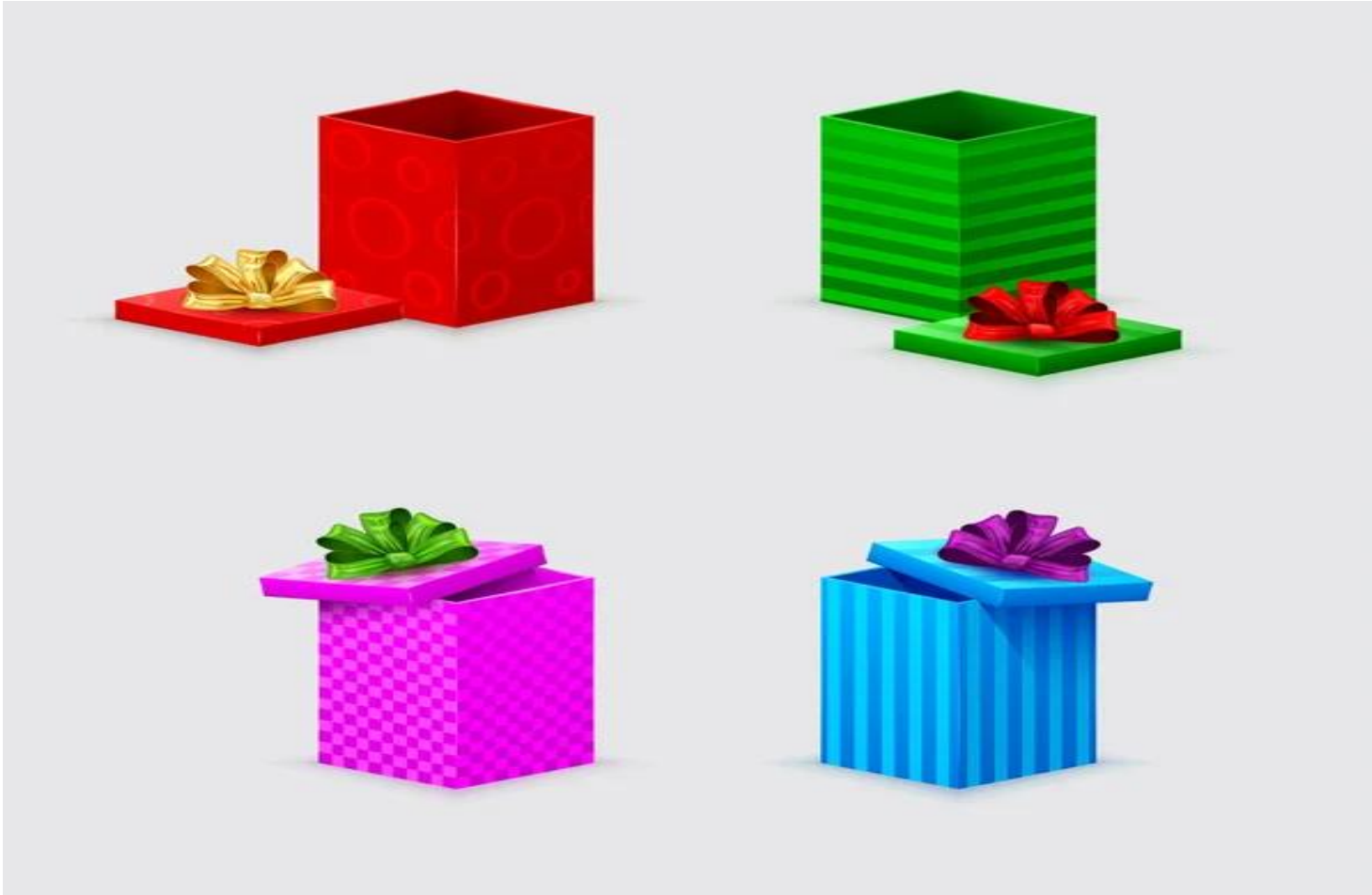
Chapter 6

More CSS Basics Key Concepts

Learning Outcomes

- Describe and apply the CSS Box Model
- Configure width and height with CSS
- Configure margin, border, and padding with CSS
- Center web page content with CSS
- Apply shadows with CSS3
- Configure rounded corners with CSS3
- Configure background images with CSS3
- Configure opacity, RGBA color, HSLA color and gradients with CSS3

Every element in a webpage is like a **physical box**, and CSS controls how this box behaves.



Width and Height with CSS

Unit	Purpose
px	Configures a fixed number of pixels as the value
em	Configures a value relative to the font size
%	Configures a percentage value of the parent element
vh	Configures a value relative to 1% of the viewport height
vw	Configures a value relative to 1% of the viewport width

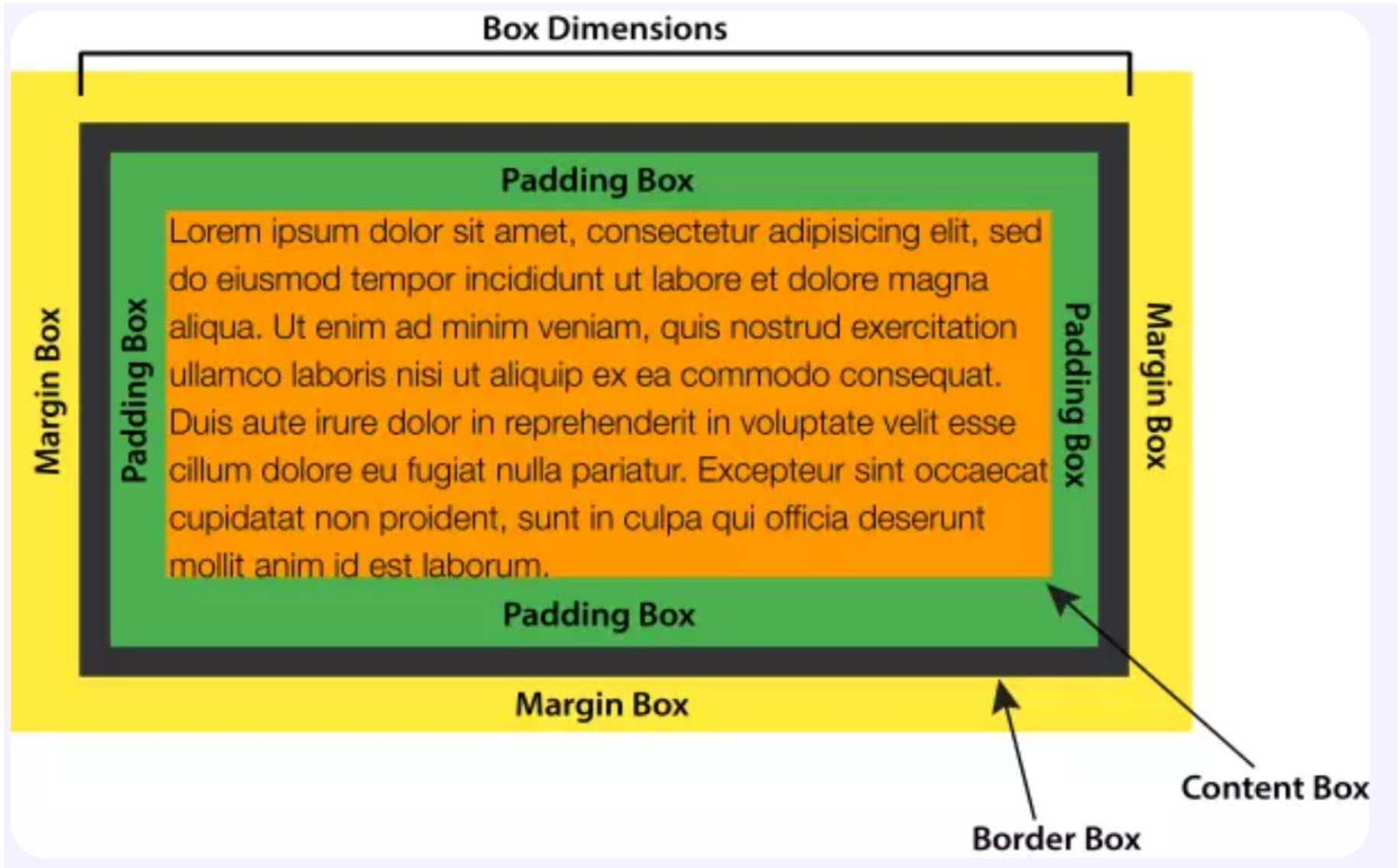
Width and Height CSS Properties

- width property
 - Configures the width of an element's content

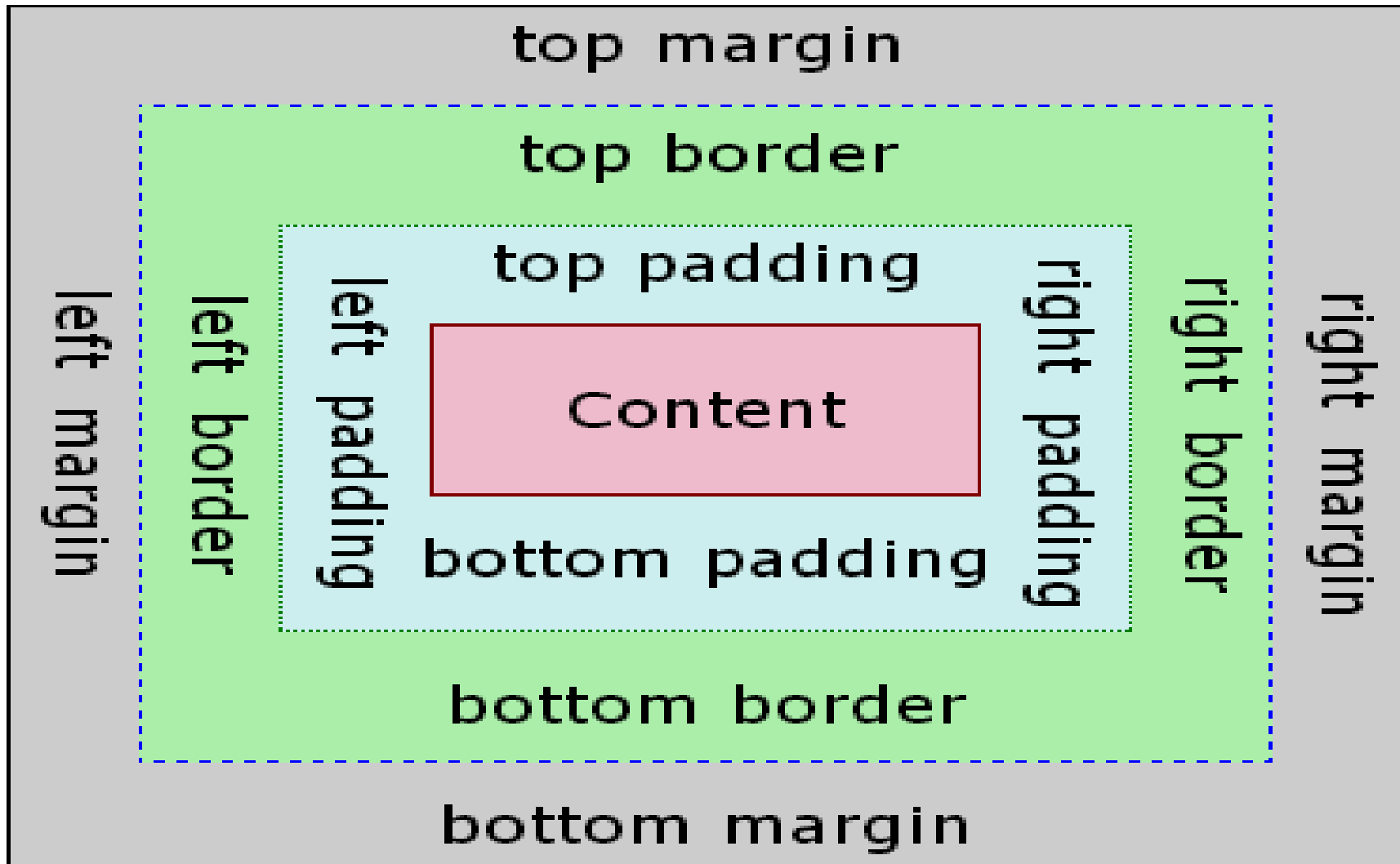
```
h1 { width: 80%; }
```
- min-width property
 - Configures minimum width of an element
- max-width property
 - Configures the maximum width of an element
- height property
 - Configures the height of an element

```
h1 { height: 100px; }
```

The Box Model



The Box Model



The Box Model

Total Width Calculation

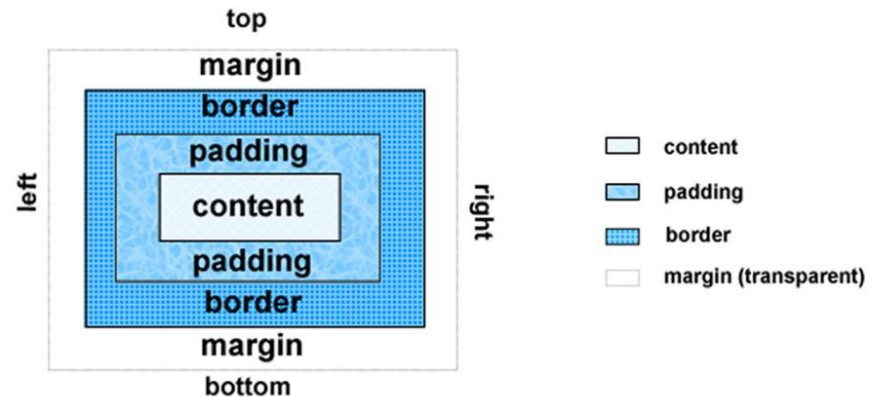
Total element width = width + left padding + right padding + left border + right border

Total Height Calculation

Total element height = height + top padding + bottom padding + top border + bottom border

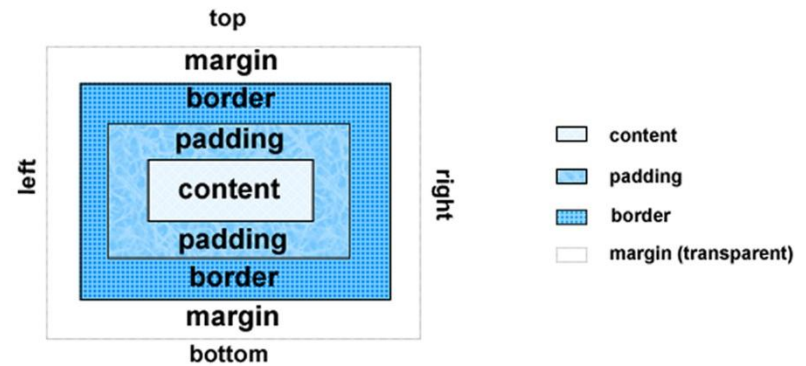
The Box Model

- **Content**
 - Text & web page elements in the container
- **Padding**
 - Area between the content and the border
- **Border**
 - Between the padding and the margin
- **Margin**
 - Determines the empty space between the element and adjacent elements

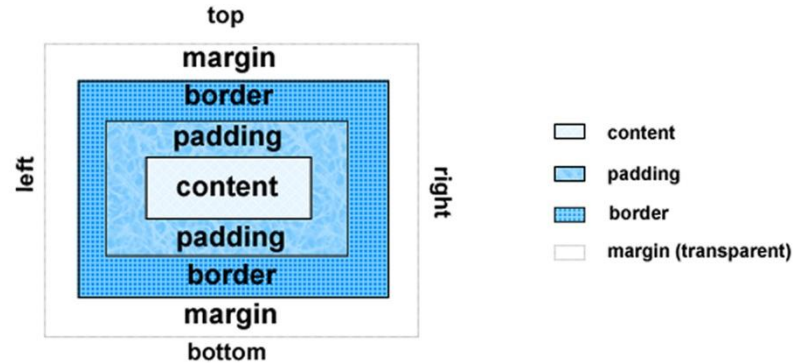


Configure Margin with CSS

- The margin property
- Related properties:
 - margin-top, margin-right, margin-left, margin-bottom

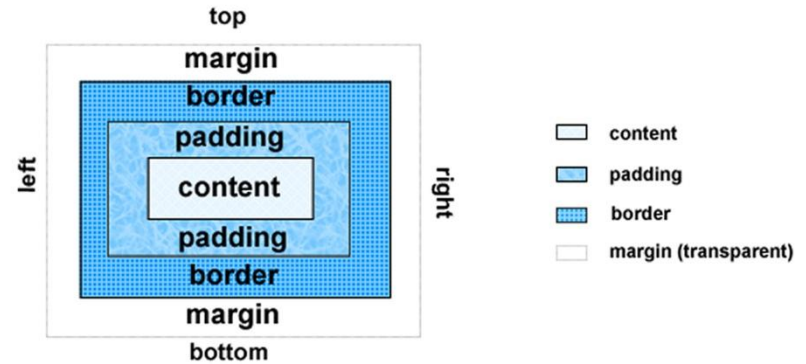


- Configures empty space between the element and adjacent elements
 - Syntax examples
 - `h1 { margin: 0; }`
 - `h1 { margin: 20px 10px; }`
 - `h1 { margin: 10px 30px 20px; }`
 - `h1 { margin: 20px 30px 0 30px; }`
- One value, like: `h1 {margin: 50px}` - all four margins will be 50px
 - Two values, like: `h1 {margin: 50px 10px}` - the top and bottom margins will be 50px, left and right margins will be 10px.
 - Three values, like: `h1 {margin: 50px 10px 20px}` - the top margin will be 50px, left and right margin will be 10px, bottom margin will be 20px
 - Four values, like: `h1 {margin: 50px 10px 20px 30px}` - the top margin will be 50px, right margin will be 10px, bottom margin will be 20px, left margin will be 30px



The CSS border Property

- Configures a border on the top, right, bottom, and left sides of an element
- Consists of
 - border-width
 - border-style
 - border-color



Heading with Border

```
h2 { border: 2px solid #ff0000 }
```

Valid border-style Values

default

none

inset

outset

double

groove

ridge

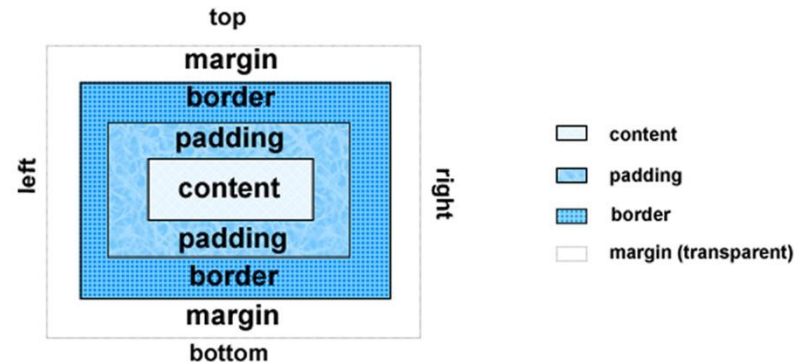
solid

dashed

dotted

Configuring Specific Sides of a Border

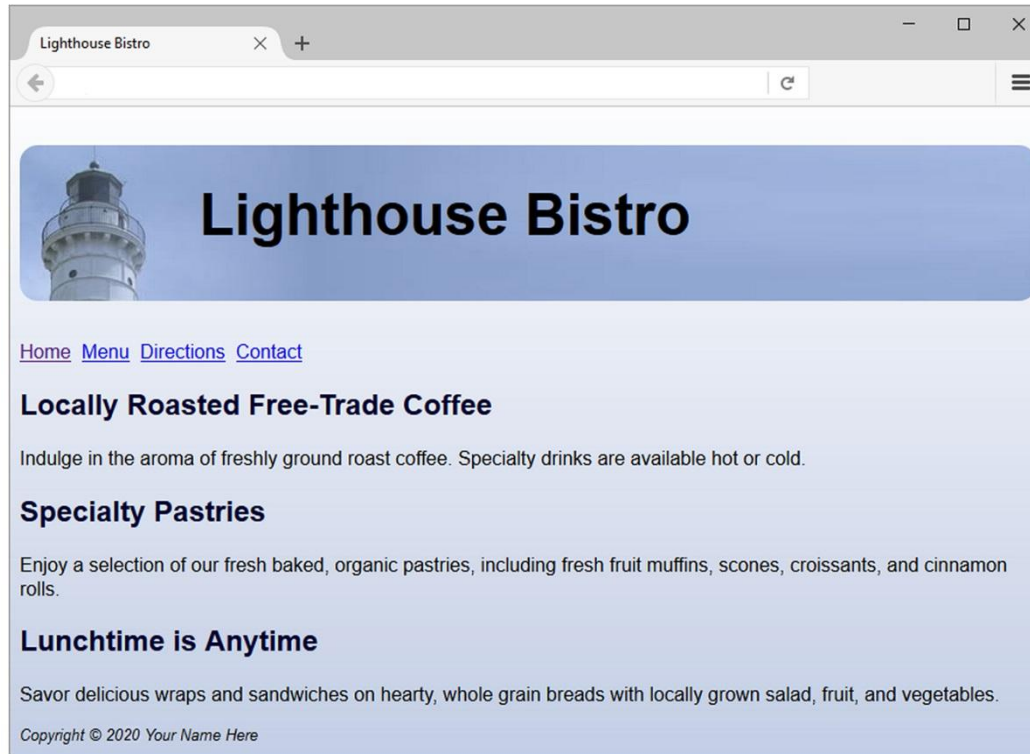
- Use CSS to configure a line on one or more sides of an element
 - border-bottom
 - border-left
 - border-right
 - border-top



Heading with Border

```
h2 { border-bottom: 2px solid #ff0000 }
```

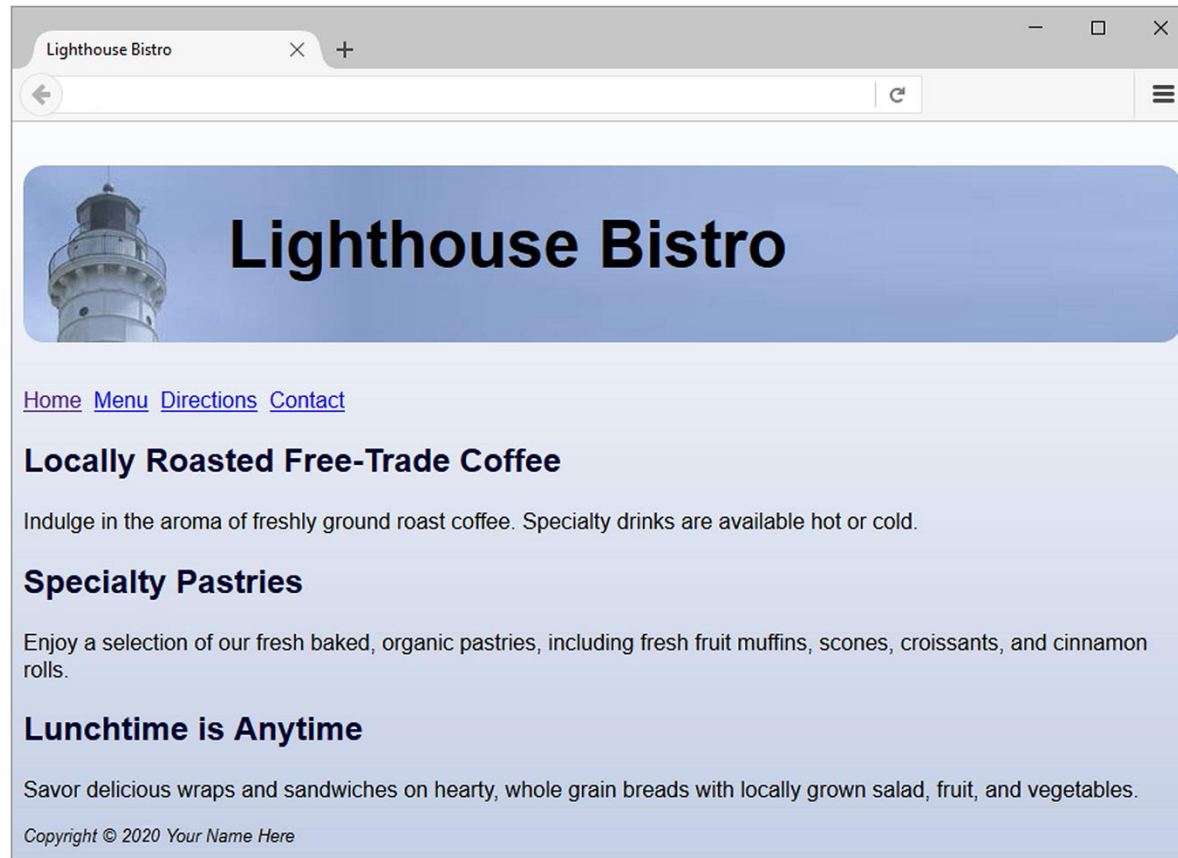
CSS Rounded Corners



- border-radius property
- Example:

```
h1 { border: 1px solid #000033;  
      border-radius: 15px; }
```

Centering Page Content with CSS



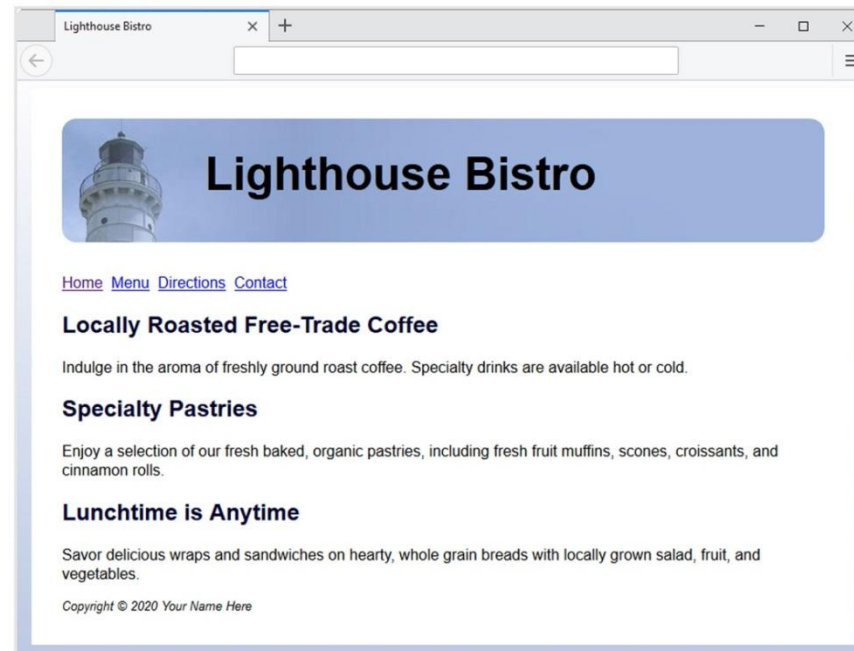
```
#container { margin-left: auto;  
              margin-right: auto;  
              width:80%; }
```


CSS3 box-shadow Property

- Configure the horizontal offset, vertical offset, blur radius, and valid color value
- Example:

```
#wrapper { box-shadow: 5px 5px 5px #828282;}
```

Note: Optional keyword: inset

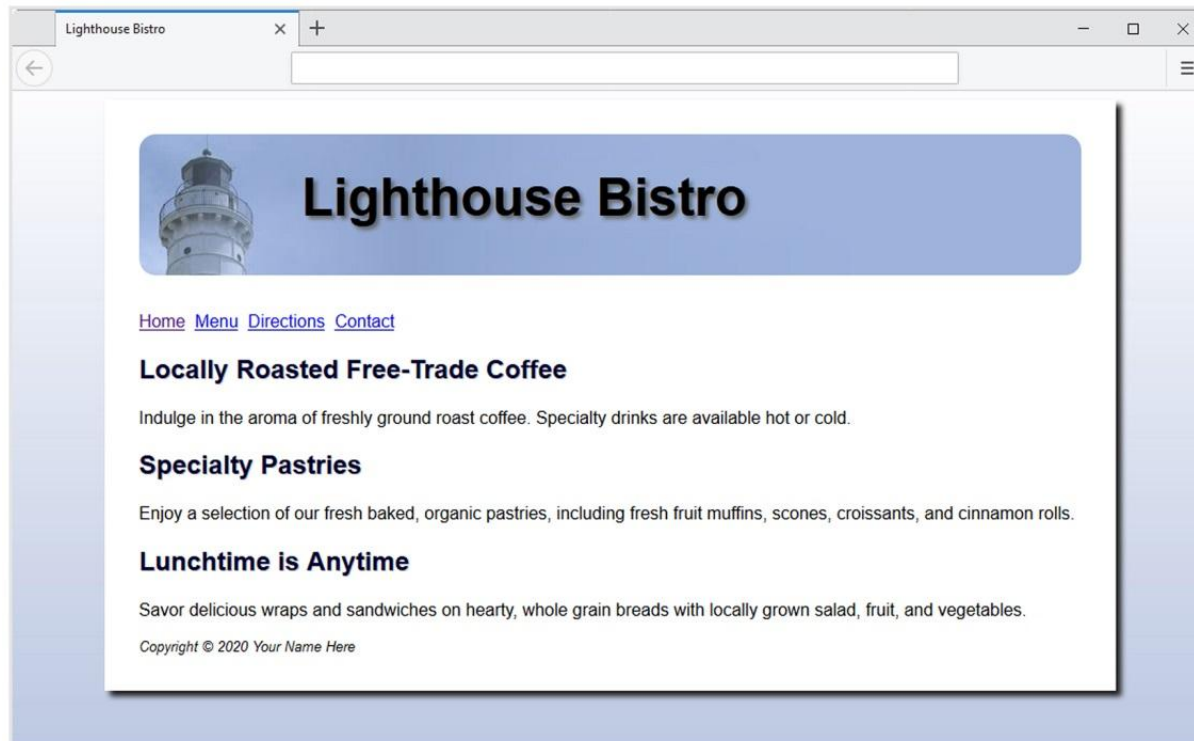


CSS3 text-shadow Property

- Configure the horizontal offset, vertical offset, blur radius, and valid color value

- Example:

```
#wrapper { text-shadow: 3px 3px 3px #666; }
```



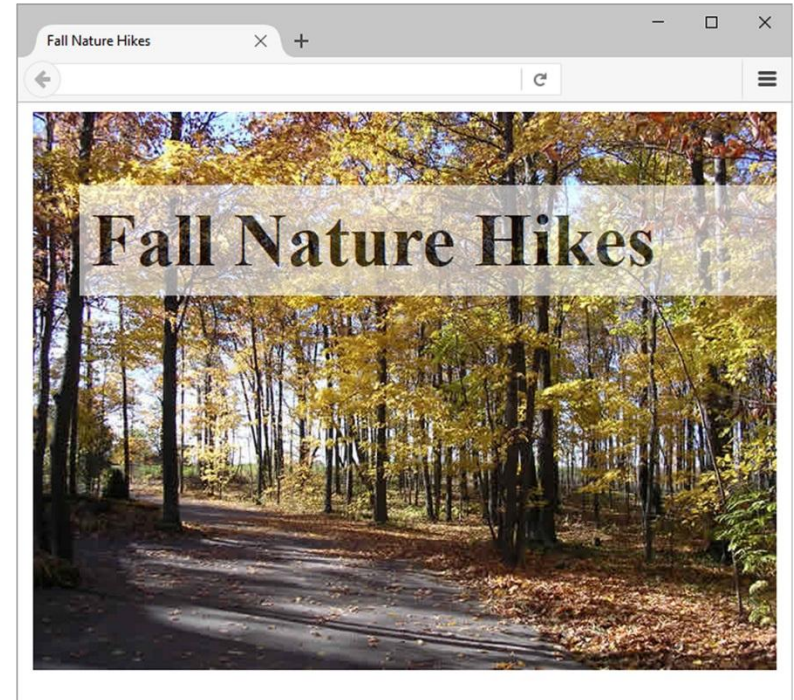
CSS Background Image Properties

- **background-clip**
 - confines the display of the background image
- **background-origin**
 - positions the background image relative to the content, padding or border
- **background-size**
 - can be used to resize or scale the background image

CSS opacity Property

- Configure the opacity of the background color
- Opacity range:
 - 0 Completely Transparent
 - 1 Completely Opaque
- Example:

```
h1{ background-color: #FFFFFFF;  
      opacity: 0.6; }
```



RGBA Color

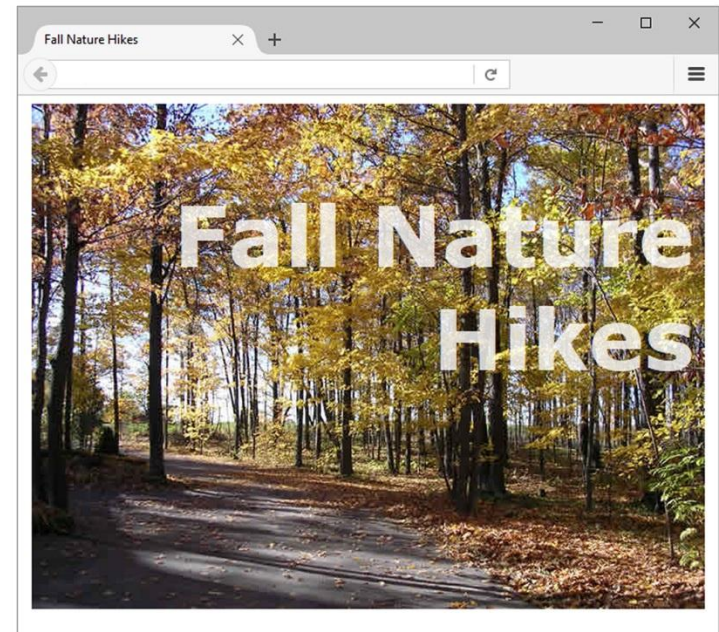
Four values are required:

- red color, green color, blue color, and alpha(transparency)
 - The values for red, green, and blue must be decimal values from 0 to 255.
 - The alpha value must be a number between 0 (transparent) and 1 (opaque).

- Example:

```
h1 { color: #ffffff;  
      color: rgba(255, 255, 255, 0.7);  
      font-size: 5em; padding-right: 10px;  
      text-align: right;  
      font-family: Verdana, Helvetica, sans-  
      serif;  
}
```

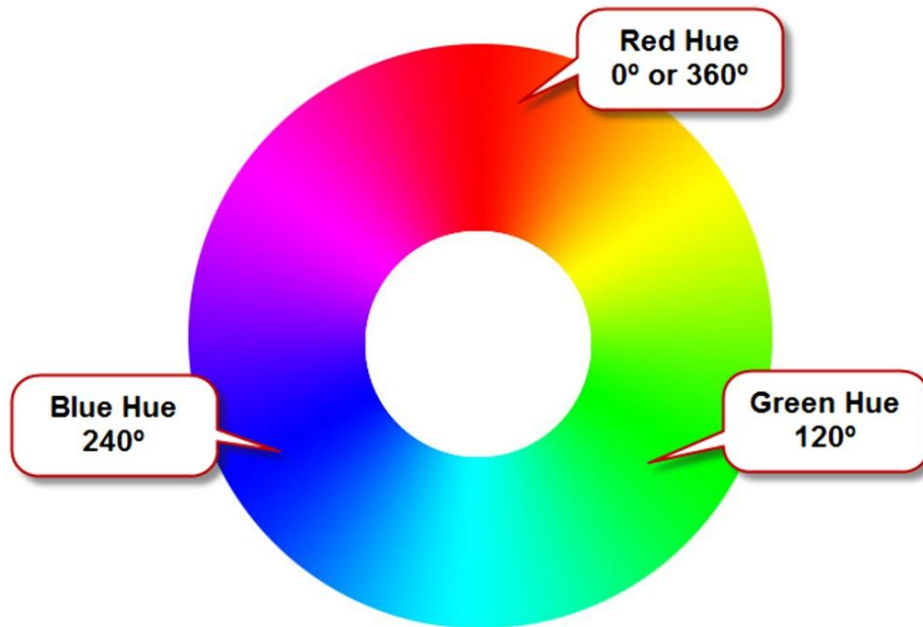
#FFFFFF rgb (255, 255, 255)	#FFFFCC rgb(255, 255, 204)	#FFF999 rgb(255,255,153)	#FFF666 rgb(255,255,102)
#FFF333 rgb(255,255,51)	#FFF000 rgb(255,255,0)	#FFCCFF rgb(255, 204, 255)	#FFCCCC rgb(255,204,204)
#FFCC99 rgb(255,204,153)	#FFCC66 rgb(255,204,102)	#FFCC33 rgb(255,204,51)	#FFCC00 rgb(255,204,0)
#FF99FF rgb(255,153,255)	#FF99CC rgb(255,153,204)	#FF9999 rgb(255,153,153)	#FF9666 rgb(255,153,102)



HSLA Color

hue, saturation, light, alpha

- Hue is a value between 0 and 360
- Saturation: percent
- Lightness: percent
- Optional alpha: from 0 to 1



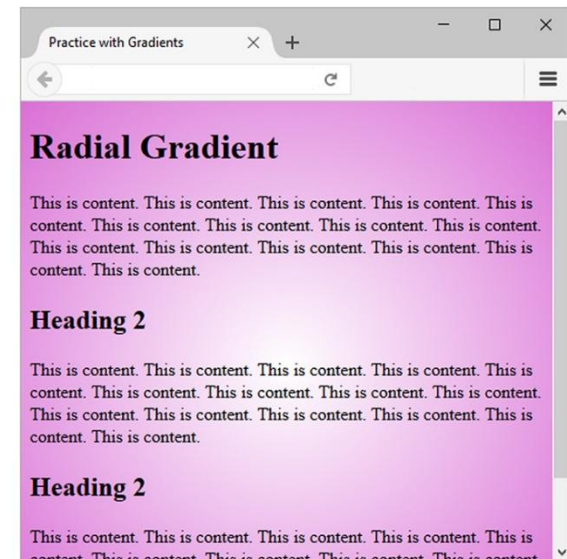
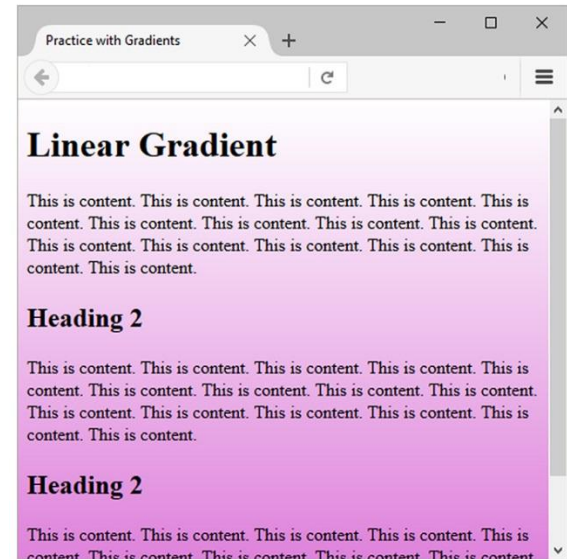
Red <code>hsla(360, 100%, 50%, 1.0);</code>
Green <code>hsla(120, 100%, 50%, 1.0);</code>
Blue <code>hsla(240, 100%, 50%, 1.0);</code>
Black <code>hsla(0, 0%, 0%, 1.0);</code>
Gray <code>hsla(0, 0%, 50%, 1.0);</code>
White <code>hsla(0, 0%, 100%, 1.0);</code>

CSS Gradients

- Gradient: a smooth blending of shades from one color to another
- Use the background-image property
 - **Linear Gradient Example**

```
body { background-color: #8FA5CE;  
        background-image: linear-gradient  
        (to bottom, #FFFFFF, #8FA5CE);  
}
```
 - **Radial Gradient Example**

```
body { background-color: #8FA5CE;  
        background-image: radial-  
        gradient(#FFFFFF, #0000FF);  
}
```



Summary

- This chapter expanded your CSS skillset.
- You were introduced to the box model.
- You configured CSS properties related to the box model, such as margin, border, padding, height, and width.
- You centered a web page using CSS.
- You explored new CSS properties including: border-radius, box-shadow, text-shadow, opacity.

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